

SNR enhancement for composite application using multiple Doppler vibrometers based laser ultrasonic propagation imager



Thanh Chung Truong, Jung Ryul Lee*

Department of Aerospace Engineering, Korea Advanced Institute of Science and Technology, Daejeon, Republic of Korea

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ABSTRACT

In recent years, the technology of using laser ultrasonic propagation imaging for damage visualization of composite structures were applied to real-world applications. Among many choices of sensor for the Ultrasonic Propagation Imager, the laser interferometry has several advantages: it is non-invasive, and portable, and with extraordinarily long-range measurement. However, the critical issue with interferometry sensing is its low signal-to-noise ratio (SNR), where the background noise can mask the damage-induced waves and making it impossible to identify the damages, especially in composite structures. In this paper, we propose a hardware-based SNR enhancement technique using multiple Laser Doppler Vibrometers (LDVs). The out-of-plane mode of ultrasonic signals are measured by multiple LDVs at a common sensing point and then averaged in real time. We showed that the SNR enhancement in experiments was consistent with the theoretical prediction, and also the test results showed a clear improvement for damage visualization of structures using Ultrasonic Wave Propagation Imaging and Ultrasonic Wavenumber Imaging algorithms.

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1. Introduction

The signal-to-noise ratio (SNR) represents the measurement of signal strength relative to background noise. This parameter has a central role in real world structural health monitoring (SHM) applications, where the main objective is the identification and quantification of damages. In the context of ultrasonic damage detection applications, the ultrasonic waves that are reflected or scattered from the damages, or are confined to the damages, can be visualized using the ultrasonic propagation imaging technique [1–9]. However, when the SNR of the measurement signal is low, the background noise masks those damage-induced waves and making it impossible to identify the damages.

The laser Ultrasonic Propagation Imager (UPI) (Fig. 1) used in this study has been developed to visualize damages in various structures [4–9]. The UPI utilizes one or more fixed sensing points and a scanning Q-switched laser beam with a high speed scan mirror. This approach is contrast with many commercial scanning LDV systems, such as the PSV-400 of Polytec, where a continuous laser beam is used for measurement of the ultrasonic waves at various locations in the scan area and a piezoelectric transducer or pulse laser is used for excitation at one fixed location. In theory, the two approaches

are equivalent due to the principle of reciprocity in linear elastodynamic medium. However, in practice, the approach chosen for the UPI has several critical advantages. First, the number of sensing points that needed surface preparation is dramatically reduced to a single point (even though for some applications, such as inspection of wind turbine blade this is still an unrealistic scenario). Second, it allows one to manufacture very high speed scanning systems thanks to the availability of high speed Q-switch lasers, high speed scan mirrors, and high speed data acquisition. Third, the UPI can be hooked up to a wide range of sensor types including contact and non-contact sensors. Finally, it is easy to extend the inspected area by using a set of electric-controlled mirrors to alter the direction of the laser beam [10]. UPI has two key limitations. First, the excitation type is often limited to single pulse excitation. Second, the focus of a high energy laser beam in an ultra-short period of time may damage the inspected specimen.

The use of laser interferometry for sensing brings remarkable advantages to the UPI system: making the system completely non-invasive, and portable, and with extraordinarily long-range measurement. However, previous studies of ours [8,9] and other authors [11,12] showed that the critical issue with interferometry sensing is its low SNR compared to contact sensors. This observation is applicable to the damage detection applications, which often utilizes the sub-micrometer amplitude laser ultrasonic waves with frequency ranging from a few kHz to a few MHz. [12] In this paper, we propose a hardware-based SNR enhancement technique that uses multiple Laser Doppler Vibrometers

* Correspondence to: Department of Aerospace Engineering, Korea Advanced Institute of Science and Technology, 291 Daehak-ro, Yuseong-gu, Daejeon 305-338, Republic of Korea.

E-mail address: leejrr@kaist.ac.kr (J.R. Lee).

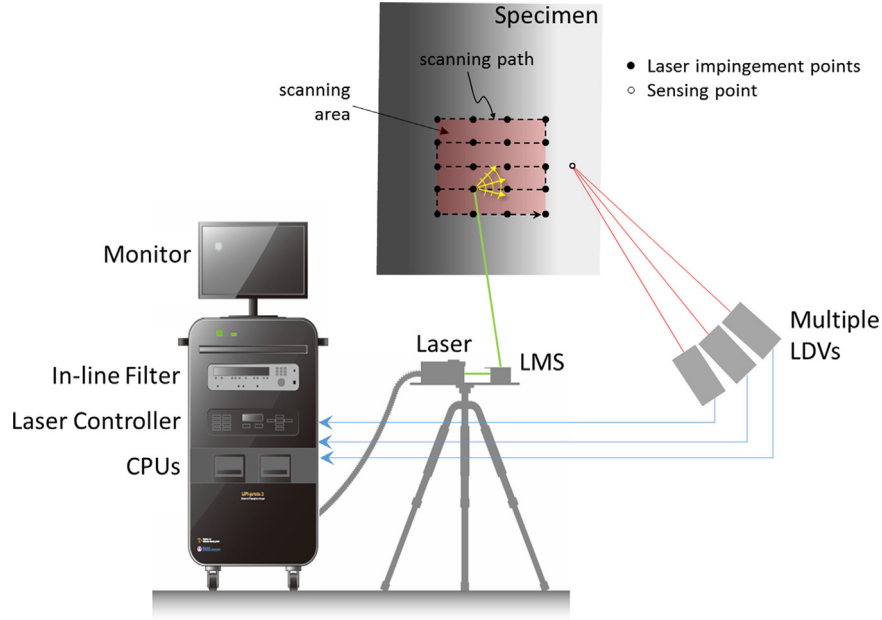


Fig. 1. Experimental setup using the multiple Laser Doppler Vibrometers (LDVs) and UPI system (LMS: laser mirror scanner, CPU: central processing unit).

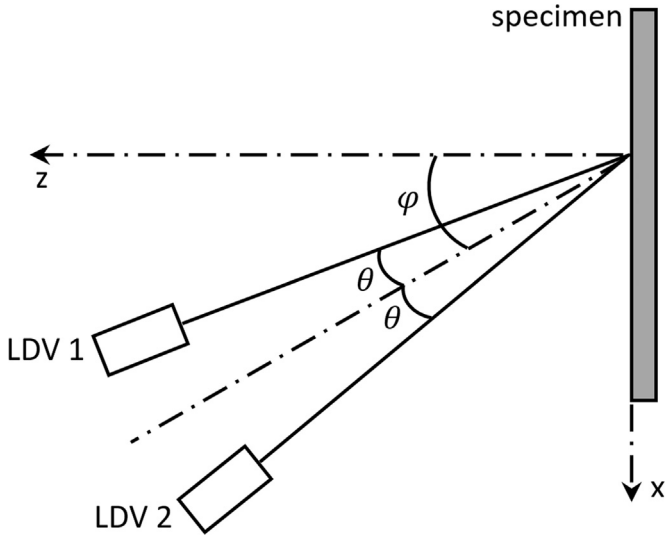


Fig. 2. Two Laser Doppler Vibrometers (LDVs) pointed at one spot on the specimen.

(LDVs) to alleviate this low SNR issue. The out-of-plane modes of ultrasonic signals are measured by multiple LDVs at one common sensing point and then get averaged in real time. The approach of utilizing multiple LDVs in this paper is differentiated from other techniques that used multiple LDVs to measure the in-plane and out-of-plane wave modes [13–15].

The paper is organized as follows. The theory and experiment to validate the proposed SNR enhancement technique are described in the next two sections. Then, brief theories of damage detection techniques of Ultrasonic Wave Propagation Imaging (UWPI) and Ultrasonic Wavenumber Imaging (UWI) are introduced. Next two damage visualization cases are presented: a 2 mm-thick carbon-fiber-reinforced-plastic wing box with impact damage, and an 8 mm-thick plastic 3D-printed object with artificial internal damage. The paper ends with a section on discussion and conclusions.

2. SNR enhancement technique for the UPI system

2.1. SNR enhancement with repeat scanning technique

In our previous paper [8], we reported the repeat scanning technique for SNR enhancement. The principle of this technique is briefly presented as follows. A mirror scanner is used to guide the Q-switch laser beam to scan a raster pattern on the target specimen. The ultrasonic wave generated at each scanning point is collected. A number of scan are then repeated by precisely controlling a scanning mirror to repeat identical scanning patterns. The signals at each scanning point are then get averaged. The SNR of the signal from one-time measurement is calculated as

$$\text{SNR} = 20 \log(A/\sigma), \quad (1)$$

where A is the signal strength and σ is the standard deviation of the noise. Under the assumption that the signal and noise are uncorrelated, and that the noise is random with zero mean and constant standard variation, the SNR of the averaged signal from n times repeat scanning is given by

$$\text{SNR} = 20 \log(n^{1/2}A/\sigma). \quad (2)$$

2.2. SNR enhancement with multiple LDVs

We will show in this section the theory of how the SNR enhancement technique with multiple LDVs works. Fig. 2 shows two Laser Doppler Vibrometers (denoted as LDV 1 and LDV 2) pointed at a common sensing point on the specimen. The z -axis is normal to the surface of the specimen, and the x -axis is on the surface of the specimen. Denote the velocities acquired by LDV 1 and LDV 2 are v_1 and v_2 , respectively. Since the line of incidence of each LDV is not perpendicular to the specimen, both in-plane (v_x) and out-of-plane (v_z) vibration modes are acquired [16]. The measured velocities of the two LDVs are

$$v_1 = v_{1(x)} + v_{1(z)} = v_x \sin(\varphi - \theta) + v_z \cos(\varphi - \theta), \quad (3)$$

$$v_2 = v_{2(x)} + v_{2(z)} = v_x \sin(\varphi + \theta) + v_z \cos(\varphi + \theta), \quad (4)$$

Assuming θ is small (the validity of this assumption will be discussed in the next section), from Eqs. (3) and (4) we have

$$v_\varphi = v_x \sin \varphi + v_z \cos \varphi = (\nu_1 + \nu_2)/2 \cos \theta \approx (\nu_1 + \nu_2)/2 \quad (5)$$

In Eq. (5), v_φ is equivalent to the measurement of an LDV at an angle φ to the z axis of the specimen. Generalizing this equation for the case of using a group of m LDVs for measurement, with the assumption that this group of m LDVs has very small differences in measurement angle, we have

$$v_\varphi \approx (\nu_1 + \nu_2 + \dots + \nu_m)/m. \quad (6)$$

A special case of Eq. (6) when $\varphi=0$, v_φ is the pure out-of-plane velocity ($v_{(\varphi=0)}=v_z$). We again assume that the signal and noise are uncorrelated, the noise of the signal is random with zero mean, and the measurements from all LDVs have the same standard variation. The last assumption is justified since, for a typical setup, the noise comes from the same source (e.g. same ground-based excitation noise) and therefore affects all of the measurements similarly. The SNR of the averaged signal measured with m LDVs and n times repeat scanning is given by

$$\text{SNR} = 20 \log(m^{1/2} n^{1/2} A/\sigma). \quad (7)$$

Eq. (7) shows that by averaging the signals from n times repeat measurement using m LDVs, the SNR (dB) of the averaged signal increases $20 \log(m^{1/2} n^{1/2})$. We will experimentally validate this observation in the next section.

2.3. Experimental validation of the proposed SNR enhancement technique

The experiment setup is shown in Fig. 3(a). Three LDVs were pointed at one sensing point on a $300 \times 300 \times 1 \text{ mm}^3$ aluminum plate. A reflective tape was attached on the aluminum plate at the sensing point to increase the amount of light reflected to the LDVs. The distance from the LDVs to the plate was 3.50 m, the distance between each LDV was 0.17 m, and the distance from the laser mirror scanner in the UPI system to the plate was 2.30 m. The 532-nm laser in the UPI system was set at a pulse repetition rate of

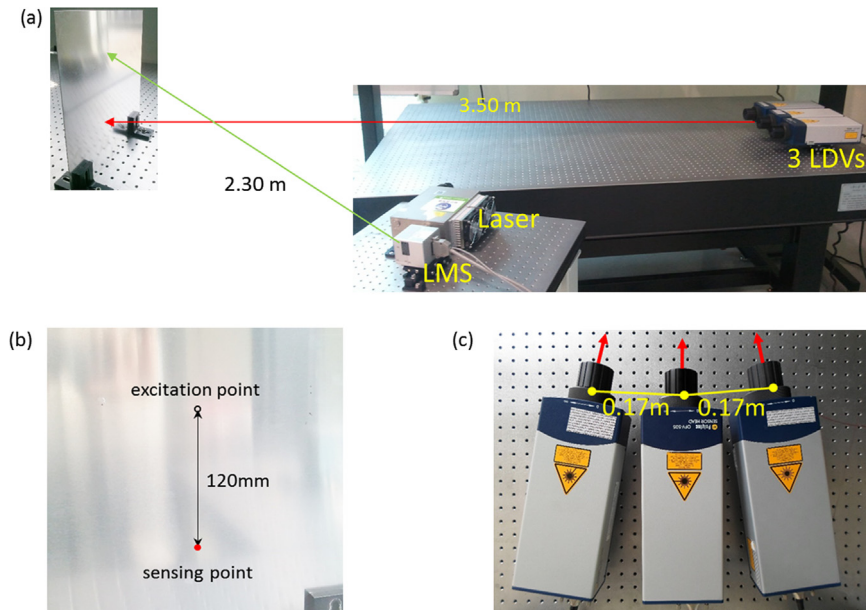


Fig. 3. (a) Experiment setup with three LDVs pointed at one spot on the aluminum plate, (b) Picture of the $300 \times 300 \times 1 \text{ mm}^3$ aluminum plate with the sensing point, and the laser excitation point at a distance of 120 mm, and (c) Picture of three LDVs, with the distance between two adjacent LDVs being 0.17 m.

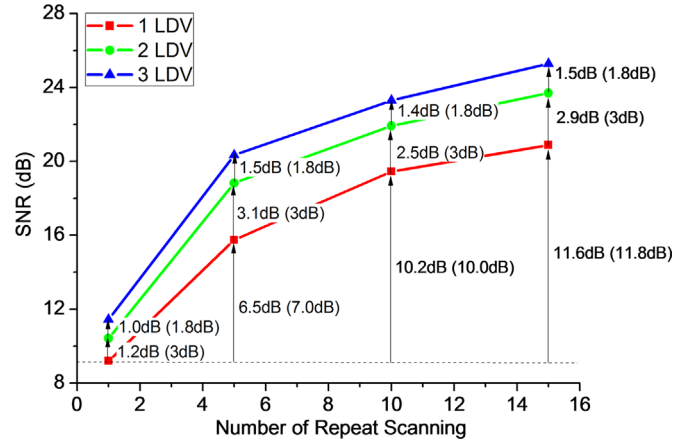


Fig. 4. Signal-to-noise ratio (SNR) comparison with different numbers of repeat scans, and different numbers of Laser Doppler Vibrometers (LDVs) for measurement. The number besides each arrow represents the actual SNR enhancement, while the number next to it and in brackets represents the theoretical SNR enhancement.

200 Hz, and the energy per pulse was 1.09 mJ. The laser excitation point and the sensing point were at a distance of 120 mm, as shown in Fig. 3(b). The collected laser ultrasonic signals from the LDV were amplified 25 times, and passed through a band-pass filter device, in the range of 30–300 kHz.

The validity of the assumption “small angle θ ” in previous section is revisited here. The error of this assumption comes from the approximation in Eq. (5) that $0.5/\cos \theta \approx 0.5$ when θ is small. The set up in Fig. 3 shows that this assumption is reasonable in real test. The distance between two adjacent LDV is 0.17 m and the distance from the LDV to the target is 3.5 m, which results in an angle θ of approximately 2.8° . The relative error then can be calculated as $(0.5/\cos 2.8 - 0.5)/0.5 \times 100 = 0.12\%$.

Fig. 4 shows the SNR comparison with using different numbers of repeated excitation, and different numbers of LDVs for measurement. The number besides each arrow represents the actual SNR enhancement, while the number next to it and in brackets represents the theoretical SNR enhancement calculated by using Eq. (7). The reasonable approximation of the theoretical prediction

to the experimental results shows the validity of our proposed SNR enhancement technique. The benefit of using a system with multiple LDVs also can be seen since the SNR of using 3 LDVs increases typically from 2 to 4 dB comparing to the case of using a single LDV. The cost associated with using a multiple LDV system is justified by two factors. First, the scanning time is reduced three fold, as we can see the SNR of the signal with single LDV with 15 times repeat scanning is the same as using 3 LDVs with 5 times repeat scanning. This factor is particularly significant for aircraft structures where the time the aircraft must be on the ground for inspection is costly. And second, as pointed by our previous publication [17], the composite structure of the aircraft is susceptible to laser pulse fatigue damage caused by the thermoelastic response to the long-term repetitive maintenance inspections. By reducing repeat scanning, we reduce the risk of damaging the structure with thermal fatigue.

3. Damage visualization algorithms UWPI and UWI

3.1. Ultrasonic Wave Propagation Imaging (UWPI)

The UPI system is utilized to scan a surface of the structure with a uniform distance between adjacent scanning points in both vertical and horizontal directions. The full scanning data forms a three-dimensional matrix $V \times H \times T$, where V and H are the number of scanning points in the vertical and horizontal directions of the scanning area, respectively, and T is the sampling length of the measurement data of each scanning point. A slide that is perpendicular to the T axis maps the ultrasonic wave propagating across of the scan area at one particular time point (Fig. 5). Successively plotting the slides along the time axis generates a video of the ultrasonic wave propagation. The damage of the structure is highlighted due to the interference effect, where the ultrasonic waves propagated directly from the laser excitation point to the receiver interfere with the ultrasonic waves reflected or confined by the damage.

3.2. Ultrasonic Wavenumber Imaging (UWI)

The UWI algorithm generates a map of estimation of the

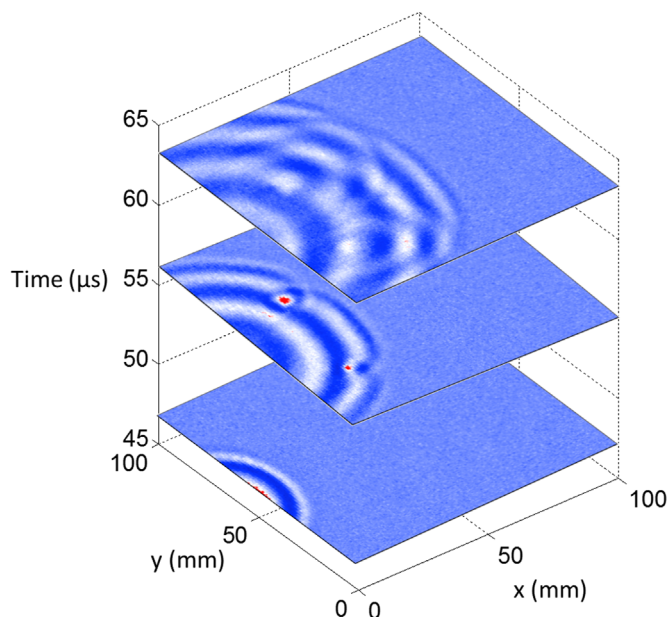


Fig. 5. Space-time wavefield of the ultrasonic wave.

ultrasonic wavenumber in the structure at a particular the wave mode and at a particular frequency. [18,19] The algorithm starts by isolating a single mode Lamb wave. First, the data in space-time domain is transformed to the wavenumber-frequency domain using a 3-dimensional (3D) fast Fourier transform (FFT). Next, a frequency-dependent band-pass wavenumber filter is applied to the wavenumber-frequency domain data. Finally, an inverse-transformed is apply to transform the data back to the space-time domain. Now, given that a single mode Lamb wave was obtained, we apply a narrowband band-pass filter in frequency domain. The result is single-mode, frequency-narrowband data in the space-frequency domain. Next, the data is passed through a bank of narrowband wavenumber filter and an inverse 3D-FFT is then applied to transform the data back to the space-time domain. The estimated wavenumber of each spatial grid point is obtained by determining the wavenumber with the highest wave energy in the bank. Since the estimated wavenumber is an indicator of local stiffness [19], the wavenumber estimation map can also be used for damage identification and characterization.

4. Applications: experiments and analysis

The proposed SNR enhancement technique was applied to the damage detection of a 2 mm-thick carbon-fiber-reinforced-plastic wing box with impact damage, and an 8 mm-thick plastic 3D-printed object with artificial internal damage.

4.1. Impact damage visualization of a 2 mm-thick carbon-fiber-reinforced-plastic (CFRP) wing box

Fig. 6 shows the top view and front view of the CFRP wing box. The scan area was $100 \times 100 \text{ mm}^2$, with the 532-nm laser set at a

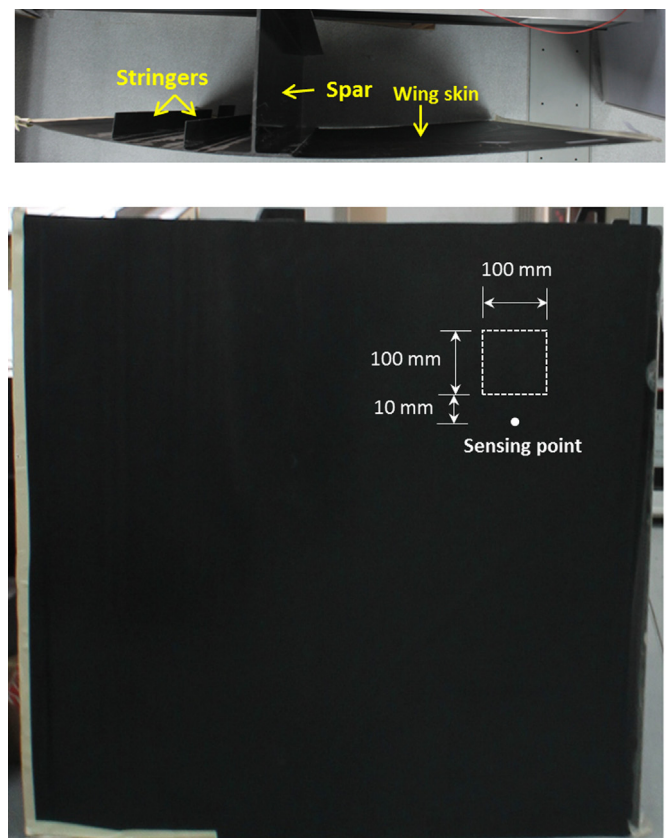


Fig. 6. CFRP wing box specimen.

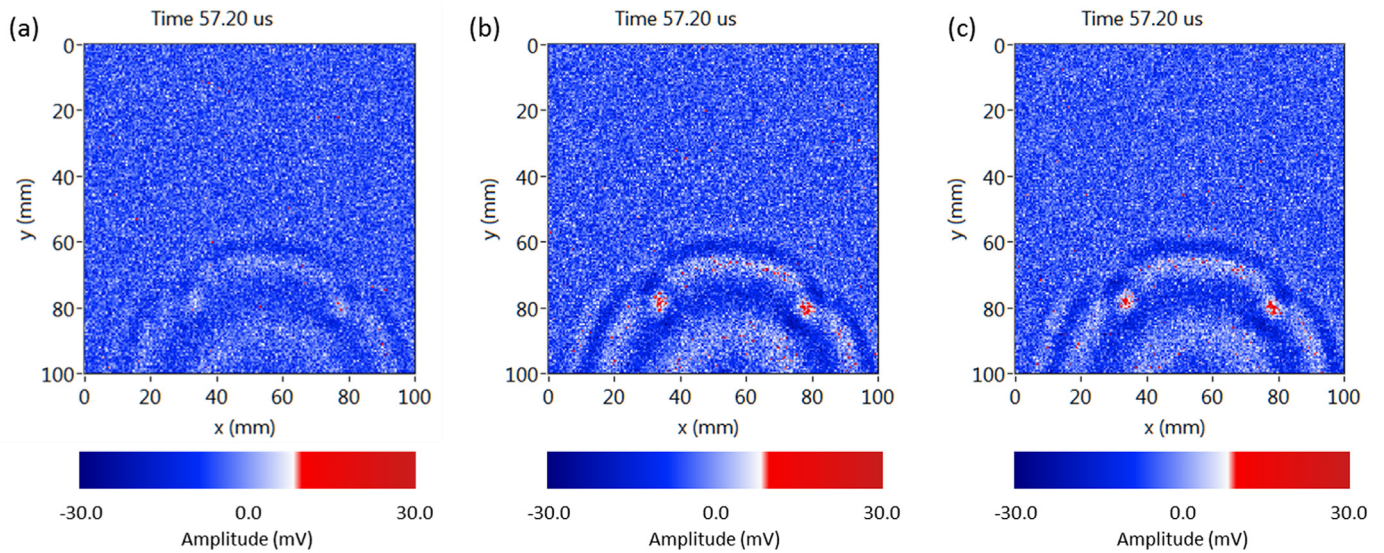


Fig. 7. Laser ultrasonic propagation image at 57.2 μ s. Results are shown using (a) 1 LDV and 5 repetitions of scanning, (b) 1 LDV and 15 repetitions of scanning, and (c) 3 LDVs and 5 repetitions of scanning.

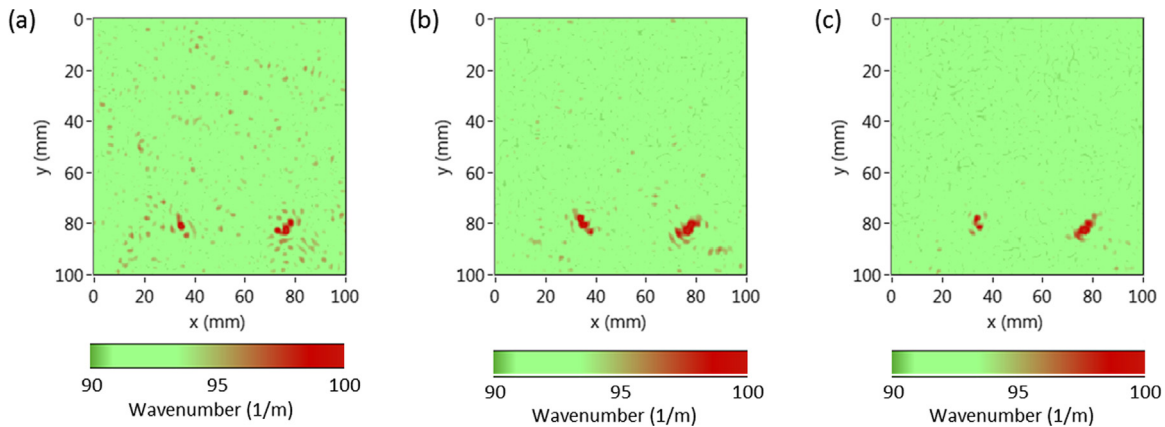


Fig. 8. Ultrasonic wavenumber estimation map. Results are shown using (a) 1 LDV and 5 repetitions of scanning, (b) 1 LDV and 15 repetitions of scanning, and (c) 3 LDVs and 5 repetitions of scanning.

pulse repetition rate of 1 kHz, and an energy per pulse of 1.20 mJ. Two impact damages were made by hitting directly to the specimen with a 15-mm diameter hammer. Those two impact damages were barely visible with the naked eye. The sensing point was at a distance of 10 mm from the lower edge of the scanning area. In order to easily achieve the same sensing point in the case of using multiple LDVs, the vibrometer sensor heads were installed on

tripods. An operator can easily adjust the LDV head manually to ensure the common sensing point of all LDVs is achieved. The collected laser ultrasonic signals from the LDVs were amplified 25 times, and passed through a band-pass filter device, in the range of 30–300 kHz. Three LDVs were pointed at one point on the wing box, with measurement lines nearly perpendicular to the surface. The distance from the LDVs to the specimen was 3.50 m, and the

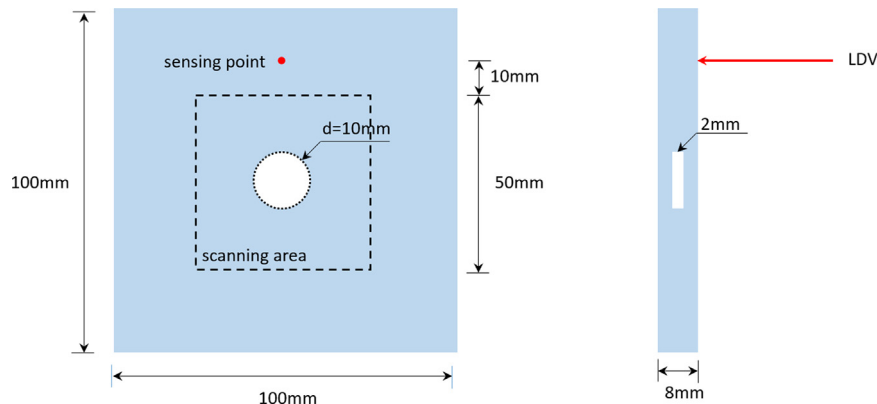


Fig. 9. The 3D-printed object (polymer material Veroblu 840) 100 mm $\times$ 100 mm $\times$ 8 mm, with a circular “damage” (acrylic compounds material SUP705) of 10 mm diameter and 2 mm thickness.

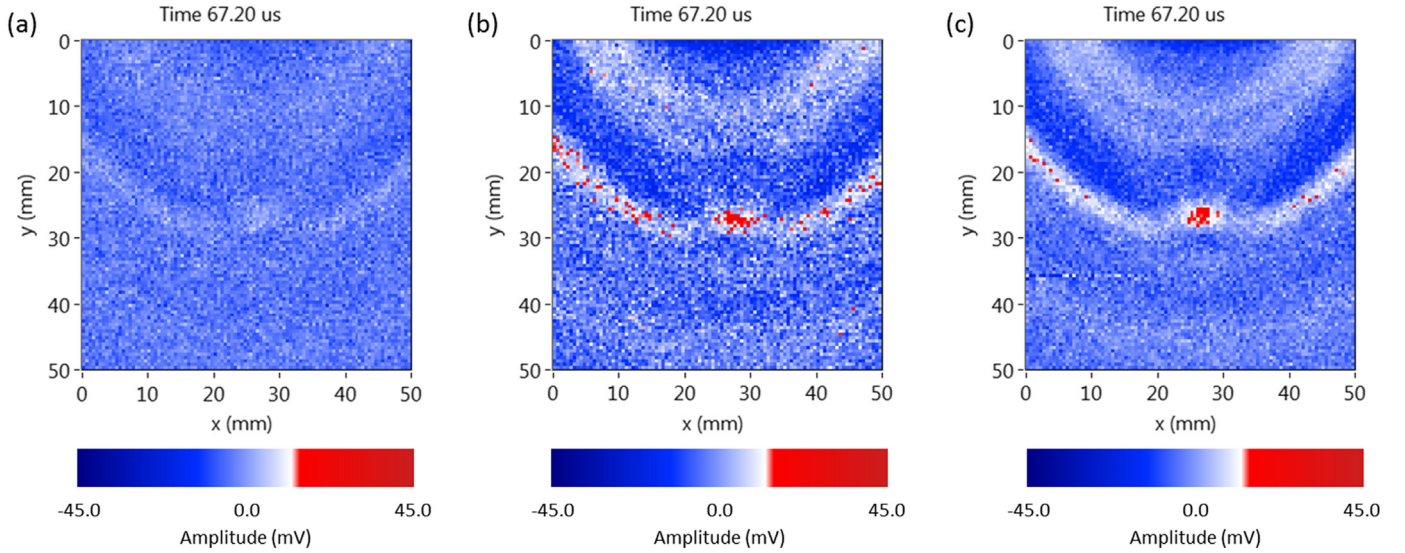


Fig. 10. Laser ultrasonic propagation image at 67.2 μ s. Results are shown with using (a) 1 LDV and 5 repetitions of scanning, (b) 1 LDV and 15 repetitions of scanning, and (c) 3 LDVs and 5 repetitions of scanning.

distance between each LDV was 0.17 m. A reflective tape was attached at the sensing point to increase the amount of light reflected to the LDVs. Fig. 7 shows the UWPI at 57.2 μ s and Fig. 8 shows the UWI maps. Three cases of using 1 LDV with 5 repetitions of scanning, 1 LDV with 15 repetitions of scanning, and 3 LDVs with 5 repetitions of scanning are presented.

4.2. Internal damage visualization of an 8 mm-thick plastic 3D-printed object

Fig. 9 shows a diagram of the 3D-printed object. The dimensions of the 3D-printed object were $100 \times 100 \times 8$ mm³ made by the polymer material Veroblu 840 with a central “damage” of 10 mm diameter and 2 mm thickness made by the acrylic compounds material SUP705. Information of the materials can be found in the manufacturer's website [20]. The scanning area was 50×50 mm². The 532-nm laser was set at a pulse repetition rate of 1 kHz, and an energy per pulse of 1.20 mJ. The sensing point was at a distance of 10 mm from the upper edge of the scanning area. The collected laser ultrasonic signals were amplified 25 times, and passed through a band-pass filter device, in the range of 30–300 kHz. Fig. 10 shows the UWPI at 67.2 μ s, and Fig. 11 shows the UWI maps. Three cases are presented similar to the previous experiment.

5. Discussion and conclusions

In this section, we will discuss the composite damage visualization results as shown in the previous section ‘Applications: experiments and analysis’. We see that the damages were successfully visualized using two algorithms UWPI and UWI thanks to the sound physical principle of each algorithm. In UWPI, the damages of the structure are highlighted due to the interference effect, where the ultrasonic waves propagated directly from the laser excitation point to the receiver interfere with the ultrasonic waves reflected or confined by the damage. In UWI, the estimated wavenumber is an indicator of local stiffness and therefore wavenumber of A_0 mode increase in case the stiffness of certain area reduces. The use of multiple LDVs for measurement improved the detectability of damages thanks to higher SNR of the measurement signals.

It is important to note the underlying assumption of this technique to ensure its proper use. First, the differences in measurement angle of multiple LDVs must be small. Second, all LDVs must be pointed to a common sensing point within the limit of a fraction of the ultrasonic wavelength. However, these conditions are not always easily obtained. These sources of error must be considered for applications with extremely long distance such as for damage detection of the wind turbine blade [21], where it is

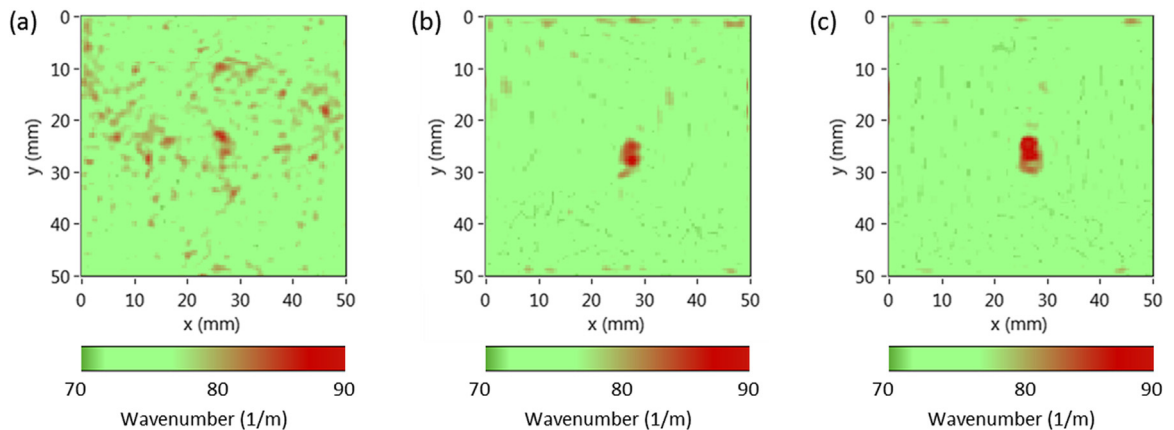


Fig. 11. Ultrasonic wavenumber estimation map. Results are shown using (a) 1 LDV and 5 repetitions of scanning, (b) 1 LDV and 15 repetitions of scanning, and (c) 3 LDVs and 5 repetitions of scanning.

virtually impossible to precisely adjust the LDV beams to have the same sensing point on the structure, for example due to wind condition that can cause the blade to vibrate. In the experiments that we presented in this paper, the distance between the LDVs and the structure is no more than a few meters with in-house environment condition. Therefore, the alignment of three LDV beams can be easily done by simple manual adjustment. Further study is needed to carefully evaluate this source of error in special cases when the misalignment is unavoidable.

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